

DIRIR AHMED-JAMA IBRAHIM

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Education

University of Ottawa

Expected Graduation: May 2028

Bachelor of Science in Computer Science (Honors)

Ottawa, ON

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Programming Paradigms, Database Systems, Operating Systems, Computer Networks

Experience

Quality Assurance Analyst Intern

Sep 2026 – Present

Senate of Canada

Ottawa, Ontario

- Developed and executed Java-based JUnit unit and functional tests to validate software functionality, identify defects, and ensure application reliability.
- Designed and executed test cases and test scenarios based on business and technical requirements to verify software enhancements and system behavior.
- Conducted User Acceptance Testing (UAT), documenting test results and defects, and validated software enhancements against business requirements.

Data Analyst Intern

May 2026 – Aug 2026

Senate of Canada

Ottawa, Ontario

- Analyzed, validated, and organized operational datasets using Excel and SQL queries, ensuring data accuracy and reliability across reporting pipelines.
- Developed Power BI dashboards and KPI reports that transformed transportation data into actionable insights, improving operational visibility for stakeholders.
- Prepared briefing notes and presented analytical findings and recommendations at the director level, supporting strategic and operational decision-making.

Projects

Net-Pulse | Go, Concurrency, Networking

- Expanded network visibility across 5 protocol layers by engineering modular DNS, TCP, port, TLS, and HTTP probes in Go.
- Reduced scan resource usage by bounding concurrent operations through a configurable worker pool, context cancellation, and per-probe timeouts.
- Validated concurrent scanning across 7 probe test suites by implementing automated unit tests, benchmarks, and race detection in CI.

CloudSentry | Python, FastAPI, AWS, Bedrock AI

- Automated AWS security auditing across 12 deterministic checks spanning IAM, S3, EC2, and CloudTrail by building a read-only scanner that never modifies live resources.
- Integrated Amazon Bedrock to generate plain-English explanations for high-severity findings, surfaced through a React/TypeScript dashboard.
- Hardened deployment security by running the scanner on EC2 with zero public inbound access, using AWS Systems Manager tunneling for all administrative connections.

PC Sentinel | Go, React, TypeScript, SQLite

- Built a Go-based Windows monitoring backend that collects CPU, memory, GPU, disk, network, and process telemetry and stores historical metrics in SQLite.
- Reduced false-positive alert fatigue by designing a deterministic rules engine with sustain periods, hysteresis, and cooldowns across warning and critical severities.
- Preserved user privacy by keeping telemetry local-only (127.0.0.1) while offering an optional AI-assisted diagnostics layer that never transmits hostnames, paths, or usernames.

Technical Skills

Languages: Java, Python, Go, C++, TypeScript, SQL, HTML/CSS

Frameworks: React, Spring Boot, FastAPI, Node.js, Express, Tailwind CSS

Cloud & Databases: AWS (EC2, S3, Bedrock), PostgreSQL, MySQL, MongoDB, SQLite

Tools: Git, GitHub, Docker, Linux, JUnit, CI/CD, L^AT_EX